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A lightweight DRR-YOLOv11s model for power equipment failure and personal protective equipment detection in hydropower stations

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The increasing demand for intelligent operation and maintenance in hydropower plants highlights the limitations of conventional inspection approaches in real-time performance and reliability, especially under edge-computing constraints. This paper presents DRR-YOLOv11s, a lightweight object detector for electrical equipment fault detection and worker safety monitoring. By optimizing the model structure for efficient inference, DRR-YOLOv11s reduces parameters by 37.58% and computational cost by 28.64% compared with the baseline, resulting in 5.88 M parameters and 15.2 GFLOPs. Experiments on the Electrical Equipment Failure dataset show a precision of 92.50%, recall of 89.42%, mAP@0.5 of 93.46%, and mAP@0.5:0.95 of 73.45%, while achieving 160.22 FPS. Cross-dataset evaluation on a Personal Protective Equipment dataset further indicates good generalization. Overall, DRR-YOLOv11s achieves a favorable trade-off between accuracy and efficiency, supporting practical deployment for real-time monitoring in hydropower plants.

Keywords Lightweight modeling, Hydroelectric plant, Power equipment failure, Personal protective equipment

Hydropower plants are fundamental infrastructure for clean energy production and play a pivotal role in promoting the transition to a low-carbon energy structure and achieving carbon neutrality goals. They also provide strategic support for grid stability and national energy security[1]. However, since hydropower plants are frequently located in mountainous or valley regions with complex terrain and limited access, their operating environments are often characterized by high humidity, strong electromagnetic interference, and seasonal load variations. Coupled with large and highly interconnected equipment systems, these conditions present challenges to traditional operation-and-maintenance (O&M) practices, such as low inspection efficiency and difficulty in fault localization. Accordingly, advanced real-time monitoring and dependable personnel safety detection technologies are required to further promote intelligent O&M in hydropower plants[2].

The existing O&M technology stack has significant limitations. Manual inspection, although a fundamental approach for obtaining equipment-status information through periodic checks, is restricted by labor costs and subjective human experience, making full-time, full-factor coverage impractical and posing safety risks under extreme conditions. Supervisory Control and Data Acquisition (SCADA) / Energy Management System (EMS) sensor networks (e.g., current, voltage, and temperature sensors) provide a data-driven monitoring backbone, but their unimodal nature limits their ability to identify visual anomalies such as surface defects and structural deformations, which are critical for effective predictive maintenance. Acoustic-based diagnostics, as a non-contact fault-detection method, can provide early warnings by analyzing operational sound signatures[3]. However, in hydropower plants, the strong background noise substantially reduces the signal-to-noise ratio, and acoustic features are highly dependent on working-condition parameters, which restricts model generalization. These limitations motivate vision-based intelligent monitoring and the integration of data-driven learning methods to enhance diagnostic precision and robustness under real-world industrial noise and illumination variations.

Recent advances in deep convolutional neural networks (CNNs) have greatly expanded the application of deep-learning-based fault detection for power equipment, substantially enhancing O&M intelligence. On the

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feature-extraction side, various hybrid architectures have been proposed. For example, He et al.[4] introduced an SA-CNN-BiLSTM framework that combines CNNs for local feature extraction, a bidirectional long short-term memory network (BiLSTM) for modeling temporal dependencies, and a self-attention (SA) mechanism to adaptively reweight features, thereby improving diagnostic accuracy and robustness. Liu et al.[5] targeted resource-constrained environments by proposing LR-DenseSENet, a lightweight model that uses spectral-map inputs and Squeeze-and-Excitation modules to retain 99.4% detection accuracy while reducing computation to roughly 20% of that of a comparable VGG network. Wang et al.[6] focused on advancing the performance-efficiency boundary of YOLOs and presented YOLOv10, a real-time end-to-end object detection model using consistent dual assignments for non-maximum suppression (NMS)-free training and a holistic efficiency–accuracy-driven design strategy, demonstrating state-of-the-art performance and efficiency with up to 46% less latency and 25% fewer parameters compared to previous models like YOLOv9. Collectively, these works underscore the potential of compact networks and multimodal fusion for practical, resource-limited industrial deployment.

Regarding object-detection algorithm evolution, mainstream approaches fall into two categories: two-stage methods based on region proposals (e.g., R-CNN family) and one-stage, end-to-end regression methods. One-stage detectors—particularly the YOLO family—offer clear advantages for real-time fault detection in power systems due to their global inference mechanisms and efficient architectures. YOLO-based methods maintain high detection accuracy while leveraging global context to reduce duplicate detections and missed detections, which is particularly beneficial for equipment-fault scenarios. Representative engineering adaptations include Lu et al.'s lightweight YOLO inspection algorithm that integrates GhostNetV2 and feature-fusion strategies, achieving 94.3% mAP and 63 FPS on a Jetson Xavier NX, demonstrating feasibility for edge deployment[7]. Dong et al.[8] proposed PHAM-YOLO for substation meter monitoring by incorporating hybrid attention and SPPF modules and optimizing localization via an EIoU loss, reporting mAP≈78.3%. Ji et al.[9] improved YOLO11's backbone with ASFF, a weighted BiFPN, and CBAM while adopting ShuffleNetV2 for lightweighting, achieving AP≈97.0 and mAP≈98.1 on insulator-defect detection, illustrating promising directions for tiny-object detection in complex scenes.

In industrial contexts, such as hydropower plants and substations, real-time detection of personal protective equipment (PPE) is essential for accident prevention. For example, Zhao et al.[10] adapted YOLOv3 for PPE detection under challenging illumination conditions. They applied gamma correction to enhance image contrast, used K-means++ to optimize anchor generation, and leveraged transfer learning for rapid adaptation. This approach yielded an mAP increase of about 2% and met real-time monitoring requirements. Zhang et al.[11] proposed the Wear-YOLO model, which enhances the perception of small targets and contextual awareness by integrating MobileViTv3, thereby improving detection robustness in complex scenes. Zhang et al.[12] proposed MEAG-YOLO, a lightweight PPE detection model for resource-constrained scenarios. It integrates MSCA, EC2f, PANet, and GhostConv, achieving a 2.4% accuracy gain while reducing FLOPs by 7.3%, thus improving recognition performance in complex substation environments.

In the field of ultra-low-power embedded safety monitoring, Papaioannou et al.[13] proposed an ultra-low-power edge-AI system for indoor environments, covering two safety-alarm tasks: fire detection and crowd counting. On the one hand, their system leverages environmental/gas-sensor measurements and employs a lightweight MLP to predict fire events. On the other hand, for crowd counting they design a lightweight YOLOv5 variant (LW-YOLOv5) inspired by ShuffleNetV2, emphasizing low memory footprint and fast inference on embedded platforms to meet real-time requirements under stringent power and computational constraints. This study is highly relevant to the edge-deployment scenario considered in this paper: under limited resources, it is essential to combine lightweight network design with efficient inference strategies to balance detection accuracy, latency, and energy consumption, thereby enabling practical real-time warnings for on-site safety monitoring.

Moreover, as the rapid advancement of edge computing technology[14] and hardware platforms continue to evolve, lightweight detection models are becoming increasingly attractive for industrial intelligent monitoring. Characterized by simple structures and low computational overhead, these models are well-suited to resource-constrained edge devices. While simultaneously maintaining high detection accuracy, lightweight models substantially improve inference speed and deployment flexibility, which is particularly important for hydropower plants with stringent real-time and energy-efficiency requirements. They can rapidly perform image recognition and provide timely alarms for equipment faults (e.g., sparks, arcs), while operating with low power consumption to reduce energy use and heat dissipation, thereby mitigating overheating-induced hardware degradation and extending device lifetime. Moreover, their low hardware dependency enables rapid deployment and replication across embedded platforms, reduces O&M costs, and supports flexible iterative updates, facilitating large-scale, multi-node hydropower monitoring systems within an "end–edge–cloud" collaborative framework[15].

Although significant progress has been made in model lightweighting[16], existing methods still face challenges in complex hydropower plant environments. Many approaches do not fully consider the unique environmental characteristics, which may affect real-time performance and lead to missed detections or false positives when detecting small-scale faults such as arcs and sparks. Furthermore, many lightweight YOLO variants (e.g., YOLOv8-lite, YOLOv10-nano, PP-YOLOE-S, and MobileYOLO) mainly focus on backbone simplification or generic operator substitution, while limited attention has been given to the joint design of backbone/neck structures and detection heads for small-scale fault patterns and PPE targets under edge constraints.

To address these issues, DRR-YOLOv11s is developed as a compact object-recognition architecture optimized for hydropower applications. It aims to improve both equipment-fault detection and worker PPE recognition, while balancing detection accuracy and inference efficiency through architecture-level design. Rather than introducing entirely new convolution or attention mechanisms, the proposed method emphasizes a structured combination and stage-wise placement strategy—"reducing computational cost in the backbone/neck, preserving feature representation through residual depthwise separable convolution blocks, and enhancing small-object sensitivity in the detection head." This design provides a coordinated optimization strategy rather

than a simple combination of existing techniques. A structural comparison with representative lightweight detectors is summarized in Table 1 to highlight the architectural characteristics of the proposed method. The key contributions are as follows:

1) A lightweight feature extraction module (DDSConv) is designed, where standard convolutions are replaced with DDSConv. This module combines depthwise separable convolutions with Ghost-based low-cost feature generation, and further integrates the SimAM attention mechanism to refine feature learning and strengthen representation capability. This design improves model performance without introducing additional parameters and with negligible computational overhead.
2) An improved detection head (RDSC). We propose a residual-enhanced depthwise separable convolution module integrated with an Effective Squeeze-and-Excitation (ESE) attention mechanism to improve small-object detection across the three detection scales (P3/P4/P5).
3) An efficient information-interaction block (RepViTBlocks). We propose an enhanced RepViTBlocks module (RVBs) that applies structural reparameterization and lightweight channel mixing to replace Bottleneck blocks, achieving lower computation while preserving representation power.
4) Comprehensive evaluation. We validate DRR-YOLOv11s on two public datasets (EEF and PPE) through ablation studies and benchmark comparisons, showing its balanced performance in terms of accuracy and computational efficiency. In addition to accuracy metrics, we report model complexity (Params, FLOPs) and runtime speed (FPS) to reflect edge-deployment readiness.

The remainder of this paper is organized as follows: Section "Methodology" presents the proposed method and systematically elaborates the architecture and optimization mechanism of the improved DRR-YOLOv11s, with an emphasis on the technical principles of the core modules, including DDSConv, RDSC, and RepViTBlocks. Section "Experiment and Results" reports the experiments and results, covering the datasets and evaluation metrics, training configurations, as well as ablation studies and benchmark comparisons for DDSConv/RDSC/RVBs. Section "Discussion" discusses the experimental results, analyzes the accuracy–efficiency trade-offs and limitations, and provides qualitative insights. Finally, Sect. "Conclusion" concludes the paper and outlines directions for future work.

Methodology

Overall network architecture design

In complex hydropower plant scenarios, safety equipment detection based on the original YOLOv11 model encounters significant limitations. Due to the constraints imposed by the convolutional receptive field and feature-fusion mechanism, the model may fail to detect small targets (e.g., sparks, arcs) or generate false detections because of complex backgrounds. To better address the resource constraints of edge-computing devices and achieve lightweight design while maintaining detection accuracy, we propose a multidimensional lightweight architecture named DRR-YOLOv11s. This model integrates three innovative components:DDSConv(Dynamic Depthwise Separable Convolution), the RDSC(Residual Depthwise Separable Convolution) detection head, and RepViTBlocks (RVBs) to jointly optimize accuracy of detection and computational efficiency. The overall network structure is presented in Fig. 1.

DDSConv lightweight fusion convolution

With the widespread application of deep learning models in intelligent surveillance, industrial inspection, and other scenarios[17], an increasing number of tasks are required to be deployed on edge devices. However, due to the limited computational resources of edge devices, reducing resource consumption and improving inference speed while maintaining accuracy have become key challenges. With the continuous enhancement of computational power and the rapid development of AI technology, the large-scale deployment of edge devices has become a clear trend. In this context, model lightweighting enables near-real-time inference on low-power devices[18].

Currently, common lightweighting techniques include structural optimization, knowledge distillation, pruning, and quantization. To tackle the high computational complexity and redundancy of traditional

Model	Backbone	Neck	Head	Lightweight strategy	Attention	Re-param
YOLOv11s(baseline)	C3k2 + SPPF + C2PSA	PAN/FPN-style	Detect(P3,P4,P5)	Compound scaling /efficient blocks	C2PSA	–
YOLOv8-lite	CSP-style lightweight blocks	PAN/FPN-style	Detect head	Channel reduction /DWConv	ECA/SE	–
YOLOv10-nano	Efficiency-driven end-to-end design	Efficiency-driven overall design	NMS-free / consistent dual assignments training paradigm	End-to-end + assignment/heads redesign	–	–
PP-YOLOE-S	CSPRepResNet backbone	PAN neck	ET-Head, anchor-free	Efficient ops + rep-style block	ESE	–
MobileYOLO	MobileNet-family backbone	lightweight neck	YOLO-style head	Depthwise separable conv heavy use	SE/ECA	–
DRR-YOLOv11s(ours)	Replace some convolutions with DDSConv and replace the C3K2/C3K bottlenecks with RVBs	Keep PAN/FPN-style fusion, but lightweight key fusion operators with DDSConv/RVBs	Insert RDSC (+ ESE) into the P3/P4/P5 detection head	Front-end cost reduction (DDSConv/RVB) + head-level compensation (RDSC)	SimAM + ESE	RepVGGDWs

Table 1. Architecture-level comparison with lightweight detectors.

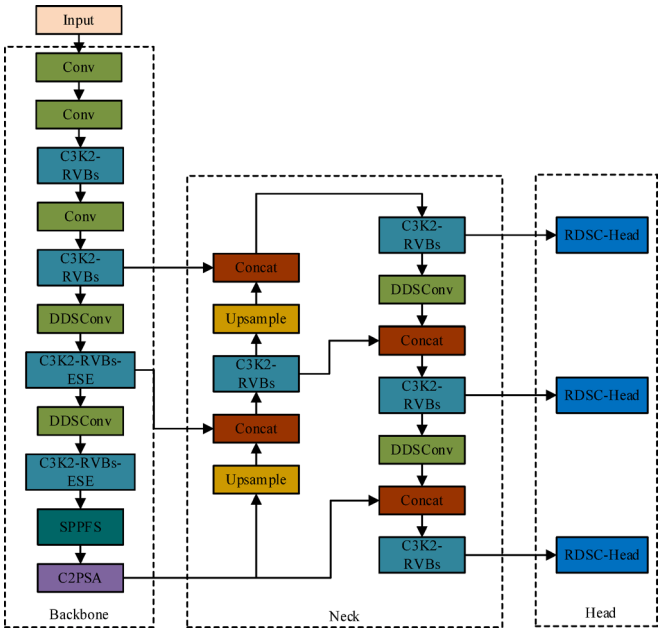


Fig. 1. Diagram of DRR-YOLOv11s network framework.

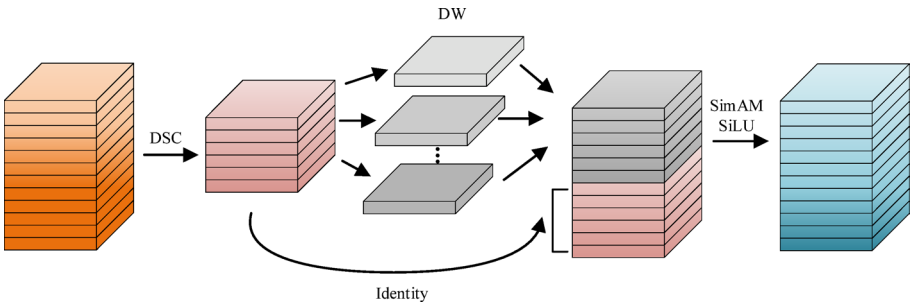


Fig. 2. Structure of DDSConv module.

convolutions, we propose a lightweight fusion convolution module named DDSConv. This module draws on the low-cost feature generation strategy of GhostNet to reduce computational costs by generating redundant low-cost features. To further improve efficiency, Depthwise Separable Convolutions (DSC) are employed to replace standard convolutions in GhostConv, achieving additional model lightweighting. Moreover, to enhance global information perception, the SimAM attention mechanism is incorporated, which improves model accuracy without increasing the number of parameters. The structure of the DDSConv module is illustrated in Fig. 2.

First, the input feature map is processed using Depthwise Separable Convolution (DSC), which was formally proposed by Google in MobileNetV1[19]. DSC consists of two steps: Depthwise Convolution (DW) and Pointwise Convolution (PW). DW performs spatial convolution independently on each input channel, largely reducing computation while preserving channel-specific information. PW then applies a 1×1 convolution to fuse cross-channel information and adjust the channel dimension without changing the spatial resolution. The parameter counts of DSC and standard convolution are given by:

$$Params_{DSC} = Params_{DW} + Params_{PW} = (C_{\text{in}} \times K \times K) + (C_{\text{in}} \times C_{\text{out}}) \tag{1}$$

$$Params_{Conv} = C_{in} \times C_{out} \times K \times K \tag{2}$$

$$\frac{Params_{DSC}}{Params_{Standard}} = \frac{(C_{in} \times K \times K) + (C_{in} \times C_{out})}{C_{in} \times C_{out} \times K \times K} = \frac{1}{C_{out}} + \frac{1}{K^2} \tag{3}$$

In Eq. (1), the first term corresponds to the parameter count of DW, while the second term corresponds to that of PW; Eq. (2) gives the parameter count of standard convolution. As indicated by Eq. (3), DSC can significantly reduce the number of parameters compared with standard convolution. For example, when $K = 3$ and typically $C_{out} \geq 64$, this ratio is approximately 1/9, meaning that the parameter count is reduced by about 88.9%.

To further reduce redundancy, we introduce DDSConv, which combines DSC with the cheap-feature generation mechanism of Ghost convolution proposed by Han et al.[20]. Different from a vanilla DSC that maps $C_{in} \rightarrow C_{out}$ directly, DDSConv first uses DSC to generate intrinsic (ghost) features with a reduced width:

$$F = DSC\left(X; C_{in} \rightarrow C_g\right), F \in \mathbb{R}^{B \times C_g \times H \times W} \tag{4}$$

$$C_g = \left\lfloor \frac{C_{out}}{r} \right\rfloor, r = ratio = 2 \tag{5}$$

Then, DDSConv produces additional cheap features using a depthwise 3 × 3 operation:

$$G = \mathrm{DW}_{3\times3}\left(F\right), G \in \mathbb{R}^{B \times C_g \times H \times W} \tag{6}$$

Finally, intrinsic and cheap features are concatenated to form expanded features:

$$E = Concat\left(F, G\right) \in \mathbb{R}^{B \times 2C_g \times H \times W} \tag{7}$$

For ease of comparison, it should be emphasized that DDSConv differs from DSC mainly in two aspects: channel mapping strategy—DSC uses PW to directly map features to C_{out}, whereas DDSConv first employs DSC to generate a narrower feature representation with C_g channels and then restores the channel width by concatenating the main-branch features with the cheap-branch features; feature generation mechanism—DDSConv introduces an additional lightweight depthwise convolution branch and applies subsequent attention enhancement, while DSC contains neither this cheap branch nor the attention refinement. Therefore, the parameter count of DDSConv (excluding SimAM, since it is parameter-free) can be expressed as:

$$\mathrm{Params}_{\mathrm{DDSConv}} = \underbrace{C_{in}K^2}_{\text{DW in DSC}} + \underbrace{C_{in}C_g}_{\text{PW in DSC}} + \underbrace{9C_g}_{\text{cheapDW}_{3\times3}} \tag{8}$$

In DDSConv, the total parameter count (excluding SimAM, which is parameter-free) is given by Eq. (8), where the first term corresponds to the DW operation in the DSC branch, the second term corresponds to the PW operation that maps C_{out} to the reduced width C_g, and the third term corresponds to the additional cheap depthwise 3 × 3 operation for generating ghost features. By dividing Eq. (8) by the parameter count of standard convolution, we obtain the reduction ratio in Eq. (9). As indicated by Eq. (9), DDSConv can further reduce the parameter count compared with standard convolution.

$$\frac{\mathrm{Params}_{\mathrm{DDSConv}}}{\mathrm{Params}_{\mathrm{Conv}}} = \frac{1}{C_{out}} + \frac{1}{2K^2} + \frac{9}{2C_{in}K^2} \tag{9}$$

For example, when K = 3 and typically C_{out}≥64 , the parameter count of DDSConv is typically no more than 7.9% of that of a standard convolution, corresponding to an approximate 92.1% reduction in parameters.

The expanded features are further refined using SimAM, a simple parameter-free attention module proposed by Yang et al.[21]. For $E \in \mathbb{R}^{B \times C \times H \times W}$, the channel-wise mean is:

$$\mu = \frac{1}{HW}\sum_{i=1}^{H}\sum_{j=1}^{W} E_{i,j} \tag{10}$$

The squared difference is:

$$E_{\mathrm{diff}}^2 = \left(E - \mu\right)^2 \tag{11}$$

The energy weight is computed as:

$$\Psi(E) = \frac{E_{\mathrm{diff}}^2}{4\left(\frac{1}{N}\sum E_{\mathrm{diff}}^2 + \lambda\right)} + 0.5, N = HW - 1 \tag{12}$$

and the final output is:

$$Y = SiLU(E \odot \sigma(\Psi(E))) \tag{13}$$

where $\sigma(\cdot)$ denotes the Sigmoid function, λ is a small constant for numerical stability, and $\odot$ indicates element-wise multiplication. SimAM introduces no additional learnable parameters, making it well-suited for lightweight optimization; SiLU is applied afterward to provide smooth nonlinearity and stable gradient flow.

RDSC module

In the lightweight decoupled detection head of YOLOv11, efficient operators such as DW/PW are commonly adopted in the classification and regression branches to reduce parameters and computational cost. However, depthwise convolution mainly models local spatial patterns independently within each channel, while cross-channel semantic dependencies are largely handled by subsequent 1 × 1 convolutions and attention-based reweighting. When targets are small, weak-textured, and easily disturbed by complex backgrounds, relying solely on lightweight local modeling may lead to insufficient discriminative cues, thereby degrading the stability and confidence of small-object detection. To this end, we propose the RDSC (Residual Depthwise Separable

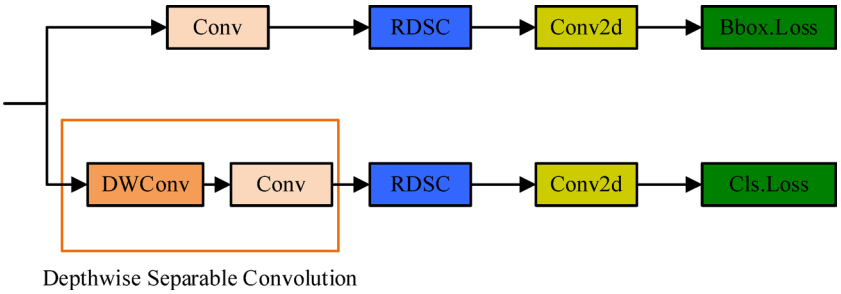


Fig. 3. Detection Head of Improved YOLOv11s.

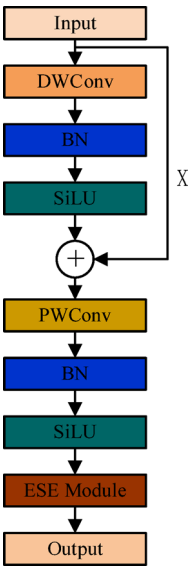


Fig. 4. Structure of RDSC module.

Convolution) module and embed it into the detection head to enhance local detail preservation, stabilize gradient propagation, and improve channel-wise saliency under lightweight constraints. As shown in Fig. 3, for the three detection scales P3/P4/P5, RDSC is inserted before the final 1×1 prediction layers in both the regression and classification branches, so that small-object features—especially on the high-resolution P3 branch—can be further strengthened.

As illustrated in Fig. 4, RDSC consists of three sequential components: a residual DW 3×3 unit (DW 3×3 + BN + SiLU with skip-add), a PW 1×1 layer (+BN + SiLU) for cross-channel fusion, and an ESE channel reweighting module. Specifically, the residual structure in the form of $Y = X + f(X)$ facilitates stable gradient propagation and preserves key details; such cross-layer connections have been demonstrated to effectively mitigate gradient vanishing and explosion issues in deep networks[22]. The subsequent PW 1×1 convolution reorganizes and fuses channel information. Finally, ESE is applied after PW to dynamically reweight channel features, highlighting channels relevant to small objects while suppressing background noise, thereby improving the detection head's sensitivity and localization stability for small targets.

The ESE module was proposed by Lee et al. at CVPR[23] as an efficient improvement over the classic SE mechanism. By optimizing the computation pipeline, ESE achieves effective channel recalibration with substantially lower overhead, making it suitable for compact model design. Its structure is shown in Fig. 5.

First, Global Average Pooling (GAP) is used to extract global statistics for each channel:

$$X_{\mathrm{GAP}} = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X(i,j) \tag{14}$$

When add_maxpool = True is enabled, Global Max Pooling (GMP) is additionally computed to capture peak activation responses:

$$X_{\mathrm{GMP}} = \max_{1 \le i \le H, 1 \le j \le W} X(i,j) \tag{15}$$

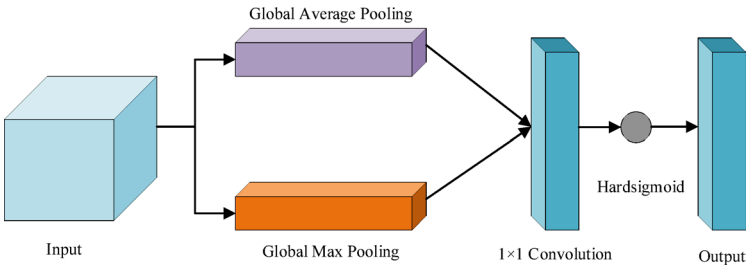


Fig. 5. ESE Module structure.

Then, GAP and GMP features are fused with equal weights (0.5:0.5), ensuring that both contribute to channel-attention computation: GAP emphasizes global distribution information, while GMP focuses on strong local responses (often more sensitive to small objects). The fused feature is transformed by a 1×1 convolution to model channel relationships and then passed through Hardsigmoid to generate gating weights for channel-wise scaling:

$$X_{\text{out}} = X \odot Hardsigmoid(W_s(0.5X_{\text{GAP}} + 0.5X_{\text{GMP}})) \tag{16}$$

where W_s denotes the learnable transformation corresponding to the 1×1 convolution and $\odot$ denotes element-wise multiplication. In summary, by combining "local-detail preservation via residual DW 3×3, cross-channel fusion via PW, and channel recalibration via ESE," RDSC enhances the feature representation capability of the detection head without reducing spatial resolution, thereby improving the accuracy and robustness of small-object detection under lightweight design constraints.

RepViTBlocks module

RepViT is a lightweight Vision Transformer (ViT) architecture proposed by Wang et al.[24] and published at CVPR. This work revisits the efficient design paradigm of lightweight CNNs by incorporating architectural concepts from lightweight ViTs, aiming to achieve high efficiency and low-latency visual processing on mobile devices. In recent years, ViT-based models such as Swin Transformer and DeiT have demonstrated advantages over conventional CNNs in tasks including image classification, object detection, and semantic segmentation. However, standard ViT models still suffer from high computational complexity, relatively weak inductive bias, and deployment challenges on resource-limited platforms. In contrast, lightweight CNN architectures are more favorable for mobile vision due to efficient GPU parallelization, stronger inductive priors, and mature deployment pipelines. By combining the global modeling capability of ViTs with the local feature extraction strength of CNNs, RepViT achieves a practical trade-off between accuracy and efficiency for mobile vision tasks.

To alleviate the excessive parameters and computational overhead of the C3K2 and C3K modules in YOLOv11, we employ an improved RepViTBlocks architecture (denoted as RVBs) to replace the conventional bottleneck structure, thereby reducing cost while maintaining detection accuracy. RVBs consists of two core submodules, namely the token mixer and the channel mixer. As illustrated in Fig. 6, the left side corresponds to the training-time multi-branch design, while the right side shows the inference-time single-branch form after structural re-parameterization. Compared with the original RepViTBlocks, our RVBs introduces three explicit modifications: enlarging the token-mixer depthwise kernels by replacing the original $(3\times3\ \text{DW})+(1\times1\ \text{DW})$ design with a $(7\times7\ \text{DW})+(3\times3\ \text{DW})$ combination to increase the effective receptive field; inserting a SiLU activation after the DW-BN operation to strengthen nonlinearity; and simplifying attention by removing the SE module inside RVBs for lower overhead, while using a lighter ESE module in high-level stages when stronger channel recalibration is needed.

The token mixer in RVBs is implemented by an enhanced RepVGGDWs module. To expand the receptive field, the original RepVGGDW is improved by replacing the 3×3 DW and 1×1 DW convolutions with a combination of 7×7 DW and 3×3 DW convolutions. In addition, SiLU is introduced after the DW-BN structure to ensure sufficient nonlinear representation, forming the enhanced RepVGGDWs. During inference, the parallel branches (7×7 DW, 3×3 DW, and identity/residual) are fused into a single equivalent convolution via structural re-parameterization, so that only one merged convolution is executed at deployment, as illustrated in Fig. 7. This design reduces branch redundancy, improves inference efficiency, and lowers latency, which is particularly beneficial for edge devices.

In this study, the SE module is omitted from RVBs to reduce unnecessary computational overhead. For the channel mixer, we adopt two 1×1 convolutional layers with BN, and insert a Gaussian Error Linear Unit (GELU) activation between them to enhance nonlinear representation capability. The entire channel mixer is further embedded within a residual structure to facilitate gradient flow and mitigate gradient vanishing.

In the shallow backbone and neck stages, RVBs is used to balance inference speed and computational cost while maintaining efficient feature extraction. For the P4 and P5 stages that primarily focus on high-level semantic representation, we replace the original SE attention with the more efficient ESE module to construct RepViTBlocks-ESE. Compared with conventional SE, ESE introduces lower computational complexity and fewer parameters while providing effective channel recalibration, enabling RepViTBlocks to achieve better performance under lightweight constraints.

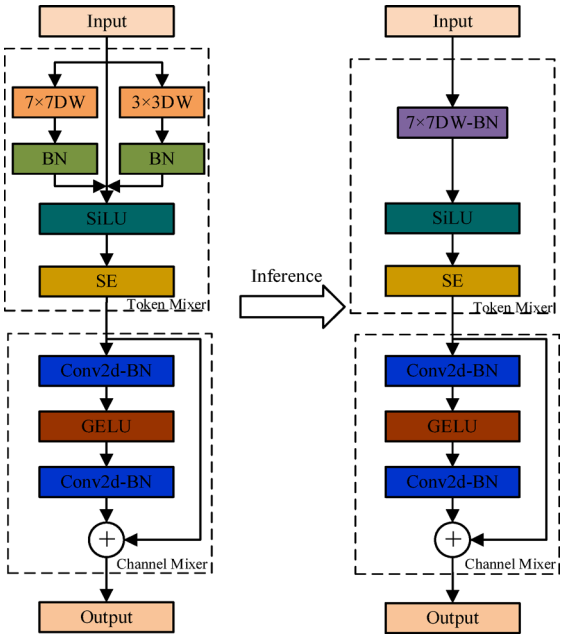


Fig. 6. Structure of RepViTBlocks module.

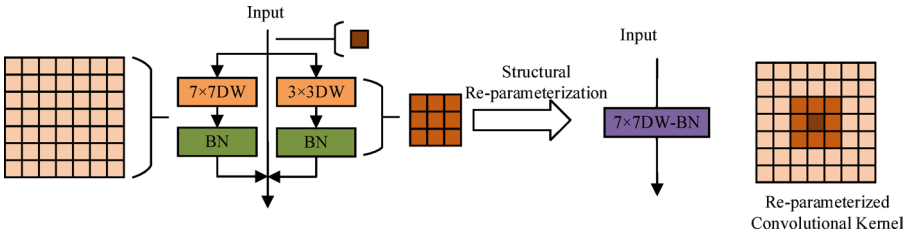


Fig. 7. Diagram of RepVGGDWs structure reparameterization process.

Name	Version model
CPU	12th Gen Intel(R) Core (TM) i7-12700F
GPU	GeForce RTX 3060
Operating system	Windows 11
Frameworks	Python-3.10.15 + PyTorch-2.1.0 + CUDA12.1
Development software	Visual studio code 2023

Table 2. Experimental environment.

Experiment and results

Experimental environment and experimental parameter settings

The setup of the experimental environment is summarized in Table 2.

Experimental dataset

During hydropower-plant operation, failure to properly wear personal protective equipment can substantially increase the risk of injury. Meanwhile, equipment faults such as arcing and sparking may cause physical damage and can further trigger severe safety incidents (e.g., fires and electric shocks). Therefore, timely and accurate detection of both equipment faults and worker PPE compliance is critical for safe and stable plant operation.

Datasets and class statistics

This study employs two publicly available datasets from the Roboflow platform. The first is the Electrical Equipment Failure (EEF) dataset, which contains 5,711 images and covers eight categories: arc, disconnector, disconnector_open, insulator, spark, switch, switchgear, and transformer. The second is the Personal Protective Equipment (PPE) dataset, which contains 8,542 images and includes five categories: gloves, helmet, person,

Class	Train Img	Train Inst	Val Img	Val Inst	Test Img	Test Inst	Total Img	Total Inst
Arc	886	1636	109	193	59	90	1054	1919
Disconnector	1045	2518	117	257	65	171	1227	2946
disconnector_open	871	1926	116	267	62	144	1049	2337
Insulator	4860	14544	635	1878	305	995	5800	17417
Spark	1197	5138	67	125	46	167	1310	5430
Switch	1175	2129	161	284	78	162	1414	2575
Switchgear	173	501	13	19	12	29	198	549
Transformer	912	1072	130	146	61	75	1103	1293
All	8571	29464	1044	3179	530	1833	10145	34476

Table 3. Class distribution of the EEF dataset.

Class	Train Img	Train Inst	Val Img	Val Inst	Test Img	Test Inst	Total Img	Total Inst
Gloves	2830	4705	898	1400	271	457	3999	6562
Helmet	2537	3789	782	1097	580	695	3899	5581
Person	3444	5275	1074	1571	675	864	5193	7710
Shoes	1875	4272	638	1399	516	1089	3029	6760
Vest	3440	5038	909	1248	630	741	4979	7027
All	5978	23079	1709	6715	855	3846	8542	33640

Table 4. Class distribution of PPE dataset.

shoes, and vest. To ensure reproducibility, Table 3 summarizes the number of images and annotated instances for each class in the augmented EEF dataset, and Table 4 reports the corresponding statistics for the PPE dataset, from which the degree of class imbalance (max/min ratio) can be directly observed.

Pre-processing and data augmentation
For all experiments, images were resized to 640 × 640. Given the relatively limited scale of the EEF dataset, offline data augmentation was applied only to the EEF training subset using Roboflow's automated tools, including horizontal flipping, brightness adjustment (± 25%), and random noise injection (0.1%), increasing the number of training images from 4,173 to 8,571. The validation and test subsets were not augmented to ensure an objective evaluation.

Train/validation/test split and evaluation protocol
Both EEF and PPE datasets were split into training/validation/test subsets with a 70%/20%/10% ratio using a fixed random seed (seed = 0) to facilitate reproducibility. Representative samples are illustrated in Fig. 8 (EEF: Fig. 8a–h; PPE: Fig. 8i–m).

Cross-dataset evaluation on PPE
To evaluate the transferability of the proposed lightweight design across different datasets, we conducted experiments on both the EEF and PPE datasets using the same DRR-YOLOv11s architecture and training configuration. Specifically, the model was first trained and evaluated on the EEF dataset, and then re-trained and tested on the PPE dataset under the same settings. The PPE results reported in this paper are obtained on the PPE test subset, serving to verify the effectiveness and stability of the proposed lightweight components under different target categories and imaging conditions.

Dataset bias and limitations
Several limitations should be noted. First, both datasets may exhibit class imbalance and long-tail categories, which can bias training toward dominant classes. Second, small or visually ambiguous objects (e.g., sparks or small insulators) are more sensitive to labeling quality and may suffer from incomplete annotations, which can lead to apparent false positives during evaluation. Third, Roboflow datasets may contain correlated samples (similar viewpoints/backgrounds), which can reduce the difficulty of the test set. These factors may affect the absolute performance values and the generalization to real hydropower sites; future work will expand multi-site data and refine annotation quality to further validate robustness.

Training settings
For all experiments, input images were resized to 640 × 640 pixels. We trained the model for 300 epochs with a batch size of 32. Stochastic gradient descent (SGD) was adopted with an initial learning rate of 0.01, a momentum of 0.937, and a weight decay of 0.0005. A linear learning-rate decay schedule (cosine disabled) was applied, where the learning rate was gradually reduced from 0.01 to 0.0001 over the entire training process. In addition, a warmup strategy was used for the first 3 epochs, with the momentum increasing from 0.8 to 0.937 and the



Fig. 8. Representative Images of Electrical Equipment Failures and Worker PPE.The human images shown in this figure are derived from publicly available datasets on the Roboflow platform.

Item	Setting
Input size	640 × 640
Epochs	300
Batch size	32
Optimizer	SGD
Initial learning rate (lr0)	0.01
LR schedule	Linear decay
Final learning rate	0.0001
Momentum	0.937
Weight decay	0.0005
Warmup	3 epochs

Table 5. Key training hyperparameters.

bias learning rate set to 0.1 during warmup. This configuration stabilizes optimization and improves detection performance. The detailed parameters are listed in Table 5.

Experimental evaluation metrics

The evaluation metrics are crucial for quantifying the performance of the assessment model. In target detection tasks, commonly-used metrics include Precision, Recall, single-threshold mean Average Precision (mAP@0.5), and multi-threshold mean Average Precision (mAP@0.5:0.95). These metrics are described in detail below.

Precision refers to the proportion of true positive cases (i.e., samples that are actually positive) among all samples predicted as positive by the model. Its mathematical formulation is presented in Eq. (17).

$$P = Precision = \frac{TP}{TP+\mathrm{FP}} \tag{17}$$

True Positive (TP) represents the number of true positive cases, and False Positive (FP) represents the number of false positive cases. This metric reflects the model's precision in predicting the positive category, that is, the share of samples correctly classified as positive among all samples predicted as positive. A higher precision indicates more reliable predictions for the positive class.

The recall rate measures the proportion of actual positive samples that are correctly detected by the model. Its mathematical formulation is shown in Eq. (18).

$$R = Recall = \frac{TP}{TP+FN} \tag{18}$$

False Negative (FN) denotes the number of false negative cases. This metric reflects the model's coverage ability in target detection, i.e., its ability to correctly identify all targets. A higher recall indicates a lower miss rate and stronger detection capability.

Average Precision (AP) measures the model's ability to maintain high precision across different recall levels by calculating the area under the Precision-Recall (PR) curve. Its mathematical formulation is given in Eq. (19),

$$AP = \int_0^1 P(R)\, dR \tag{19}$$

P(R) denotes the precision at a given recall rate R. Here, P(R) represents the precision at recall rate R. AP is typically computed based on detection results at various Intersection over Union (IoU) thresholds. IoU quantifies the degree of overlap between the predicted and true bounding boxes, and its mathematical expression is shown in Eq. (20),

$$IOU = \frac{|A \cap B|}{|A \cup B|} \tag{20}$$

A denotes the area of the ground-truth bounding box, and B denotes the area of the predicted bounding box. A higher IoU value indicates a better alignment between the predicted and actual locations, resulting in improved detection performance.

Mean Average Precision (mAP) is the arithmetic mean of the AP values across all target categories. Its mathematical formulation is shown in Eq. (21),

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i \tag{21}$$

AP_i denotes the average precision of the ith category, and N represents the total number of categories. To comprehensively evaluate both localization and classification performance, the mAP@0.5 metric is introduced, which represents the mean Average Precision when the IoU threshold is set to 0.5. Specifically, if IoU > 0.5, the predicted bounding box is considered to correctly contain the target; if IoU < 0.5, the prediction is considered invalid. The calculation is shown in Eq. (22).

$$mAP@0.5 = \frac{1}{N} \sum_{i=1}^{N} AP_i\,(IoU = 0.5) \tag{22}$$

Furthermore, mAP@0.5:0.95 evaluates the model's performance under varying localization accuracy requirements by averaging the mAP over IoU thresholds ranging from 0.5 to 0.95, with a step size of 0.05.

Additionally, this paper comprehensively evaluates the optimization performance of models through metrics including model parameter count, computational complexity, and frames per second (FPS). Among these, FPS serves as a critical indicator for measuring the processing speed of object detection algorithms, defined as the number of image frames an algorithm can process per second. Its value directly reflects the model's operational efficiency and real-time detection ability.

Analysis of experimental results

Confusion matrix

The confusion matrix used to evaluate the performance of the DRR-YOLOv11s model is shown in Fig. 9.

The confusion matrix illustrates how the detector assigns predicted instances to different categories in the target-detection task. Specifically, it summarizes the correspondence between predicted labels and ground-truth labels under a fixed operating point, thereby providing an intuitive view of common confusions. In this work, the confusion matrix is constructed after NMS by matching predicted boxes to ground-truth boxes using a one-to-one assignment with an IoU threshold of 0.5, and retaining predictions whose confidence score is ≥ 0.5. Each entry therefore reflects the distribution of matched predictions at this threshold setting (i.e., a threshold-dependent analysis), rather than a threshold-free detection metric.

As shown in Fig. 9, at a confidence threshold of 0.5, the disconnector_open class exhibits a high proportion of correctly matched predictions (approximately 98%), indicating a strong class-wise performance at this operating point. The switchgear, disconnector, and transformer classes also show similarly strong behavior, with correct-match proportions of around 97%, 95%, and 95%, respectively. It should be emphasized that the confusion matrix is provided for qualitative interpretation under a fixed threshold, whereas the primary quantitative evaluation of detection performance is reported using class-wise AP (and mAP).

Some categories show relatively lower performance at this operating point. For example, arc and spark achieve correct-match proportions of approximately 89% and 83%, respectively. These targets often appear as small-scale luminous points, which increases detection difficulty and makes them more sensitive to confidence thresholding. The insulator class shows a correct-match proportion of about 88%; meanwhile, a relatively high rate of false positives on background is observed. This is mainly attributed to incomplete annotations in the dataset (e.g., small insulators that were not labeled), further aggravated by blurred viewpoints and non-uniform illumination, which increases the likelihood of spurious detections. Overall, the DRR-YOLOv11s model demonstrates stable detection behavior, and its final performance is mainly supported by the reported class-wise AP metrics.

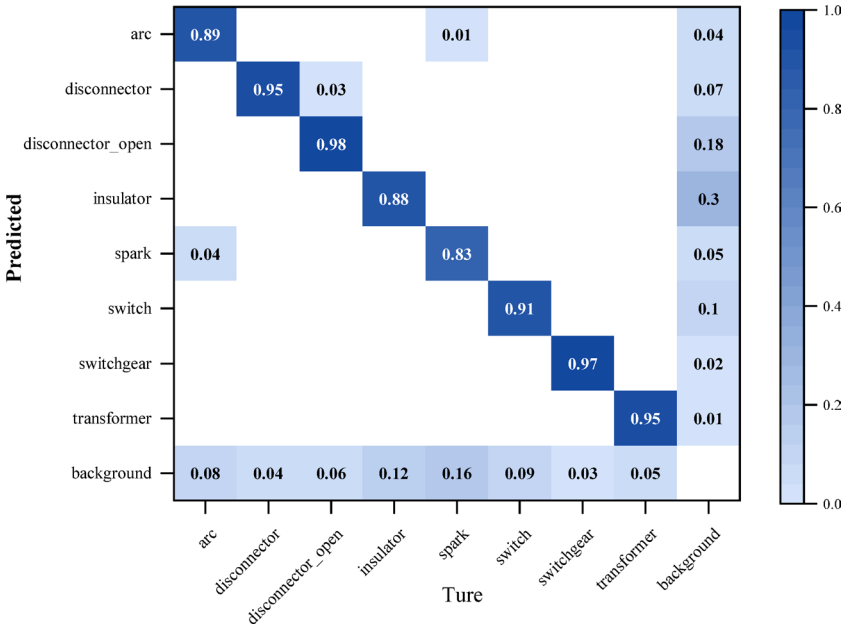


Fig. 9. Confusion matrix.

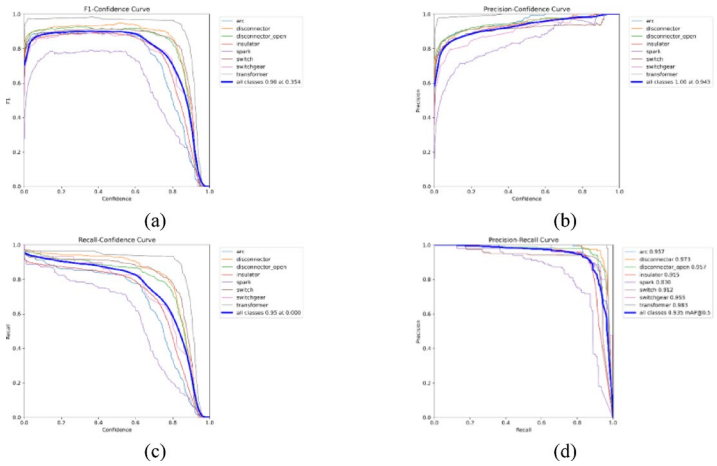


Fig. 10. Performance curves.

Performance curve

In the field of image target detection, there exists a strong intrinsic correlation among the F1 score, Precision, and Recall. These interrelated evaluation metrics offer a comprehensive approach to assessing and optimizing model performance. By analyzing these metrics, one can gain an in-depth understanding of the model's strengths and weaknesses and implement targeted strategies to enhance its performance. The performance curves of the DRR-YOLOv11s model are presented in Fig. 10.

Figure 10a illustrates the relationship between the confidence level and the F1 score. As shown, the F1 score reaches its maximum within the confidence interval of 0.2–0.6, and the curve is approximately linear in this range. Notably, the F1 score of the spark category is below the average, mainly because the target size in the dataset for this category is relatively small, which increases the likelihood of missed detections. The curve indicates that precision and recall are better balanced within this interval. To achieve optimal equipment fault detection, the confidence level is typically set around 0.5.

Figure 10b depicts the relationship between precision and the confidence level. As the confidence level increases, precision rises, while recall decreases. Under medium-confidence conditions, the precision of each category reaches approximately 0.9, demonstrating the model's strong overall detection performance.

Figure 10c shows the relationship between recall and the confidence level. In contrast to precision, recall gradually decreases as the confidence level increases. However, within the 0.4–0.6 confidence interval, recall

remains around 0.9, indicating that the model maintains excellent target detection capability and effectively minimizes missed detections.

Figure 10d presents the precision-recall curve. The curve is close to the upper-right corner, and mAP@0.5 reaches 0.935, indicating that the model achieves excellent comprehensive detection performance, accurately detecting targets while maintaining a high recall rate.

Experimental training results

The training results of DRR-YOLOv11s are shown in Fig. 11.

When the number of training epochs is less than 100, the loss function decreases rapidly, while the precision, recall, and average precision increase sharply. This indicates that the model converges effectively at this stage and rapidly learns the relevant features from the data. During the period from 100 to 200 epochs, the rate of loss reduction slows down significantly, and the growth of precision, recall, and average precision starts to stabilize. This suggests a slower learning rate, with smaller parameter adjustments as the model gradually approaches an optimal solution.

After 200 epochs, the loss curves on the validation set tend to stabilize with no significant fluctuations, and other performance metrics also show smooth trends. This demonstrates that the network has converged; the model's learning process from the training data has reached a stable state, and parameter changes are minimal. In addition, although the validation loss is slightly above the training loss, the small difference demonstrates strong generalization and no evident overfitting.

Parameter count comparison

The comparison results of the number of parameters between YOLOv11s and DRR-YOLOv11s are shown in Table 6.

The table illustrates that DDSConv decreases the number of parameters for various input–output channel combinations. Specifically, when the input channels are 256 and output channels are 512, the number of parameters decreases from 1,180,672 to 70,144, corresponding to a reduction of 1,110,528, or 94.06%. When both input and output channels are 256, the model achieves a reduction of 554,112 parameters, equivalent to 93.86%. For 128-channel configurations, the parameter count is lowered by 137,792, a 93.28% decrease. These results indicate that DDSConv can substantially compress model parameters with minimal loss of accuracy, demonstrating excellent lightweight performance in both high- and low-channel scenarios.

The C3K2-RVBs module also exhibits strong parameter optimization capabilities. For example, with 756 input channels and 512 output channels, the number of parameters is reduced from 1,511,424 to 1,069,056, a reduction of 442,368, or 29.27%, representing the largest reduction among the module's configurations. Its improved version, C3K2-RVBs-ESE, when both the input and output channels are set to 512, also experiences a parameter reduction of 409,344, representing a 29.66% decrease. Other configurations maintain a relatively stable reduction of around 29%, indicating that both C3K2-RVBs and C3K2-RVBs-ESE achieve nearly 30% parameter compression across various channel settings, reflecting excellent model lightweighting performance.

Furthermore, the detection head module has been optimized for lightweighting, reducing the parameter count from 821,343 to 787,167, a reduction of 34,176 or 4.16%.

Comparison of effective receptive fields of the model

To quantitatively and precisely evaluate the model's Effective Receptive Field (ERF), this study adopts the visualization method proposed by Ding et al.[25], which represents the model's ability to perceive features at different scales through image and data analysis. Key feature layers of YOLOv11s and its improved version, DRR-YOLOv11s, were selected for ERF characterization. Attention was given to four layers: the backbone's 10th layer and the Neck output layers P3, P4, and P5 for small, medium, and large target detection.

The ERF distribution characteristics of each layer were assessed using the Max-Value metric and the proportion of multi-threshold contributing regions, with specific threshold values including 20%, 30%, 50%, 99%. The Max-Value metric reflects the sensitivity of the model's response to the central pixel. As shown in

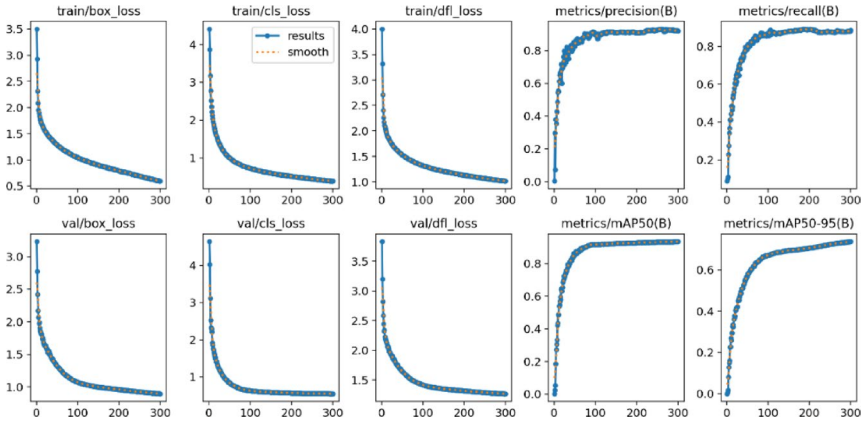


Fig. 11. Training result graph of DRR-YOLOv11s.

YOLOv11s	Parameters	DRR-YOLOv11s	Parameters
Conv [3,32]	928	Conv [3,32]	928
Conv [32,64]	18,560	Conv [32,64]	18,560
C3K2 [64,128]	26,080	C3K2-RVBs [64,128]	23,040
Conv [128,128]	147,712	Conv [128,128]	147,712
C3K2 [128,256]	103,360	C3K2-RVBs [128,256]	87,040
Conv [256,256]	590,336	DDSConv [256,256]	36,224
C3K2 [256,256]	346,112	C3K2-RVBs-ESE [256,256]	247,936
Conv [256,512]	1,180,672	DDSConv [256,512]	70,144
C3K2 [512,512]	1,380,352	C3K2-RVBs-ESE [512,512]	971,008
SPPF [512,512]	656,896	SPPF [512,512]	656,896
C2PSA [512,512]	990,976	C2PSA [512,512]	990,976
Upsample + Concat	0	Upsample + Concat	0
C3K2 [768,256]	443,776	C3K2-RVBs [768,256]	370,176
Upsample + Concat	0	Upsample + Concat	0
C3K2 [512,128]	127,680	C3K2-RVBs [512,128]	111,360
Conv + Concat [128,128]	147,712	DDSConv + Concat [128,128]	9920
C3K2 [384,256]	345,472	C3K2-RVBs [384,256]	271,872
Conv + Concat [256,256]	590,336	DDSConv + Concat [256,256]	36,224
C3K2 [768,512]	1,511,424	C3K2-RVBs [768,512]	1,069,056
Head-Detect	821,343	Head-Detect-RDSC	787,167

Table 6. Comparison of the number of parameters between YOLOv11s and DRR-YOLOv11s.

Model	MaxValue	t = 20% (%)	t = 30% (%)	t = 50% (%)	t = 99% (%)
YOLOv11s(10)	0.37	4.58	7.82	17.67	97.82
DRR-YOLOv11s(10)	0.46	4.72	7.82	17.14	97.21
YOLOv11s(16)	2.55	0.49	1.23	5.87	95.98
DRR-YOLOv11s(16)	3.30	0.74	1.68	6.17	94.15
YOLOv11s(19)	0.91	1.45	2.80	8.72	96.59
DRR-YOLOv11s(19)	1.09	2.30	3.94	10.46	95.98
YOLOv11s(22)	0.32	6.81	11.71	23.31	98.44
DRR-YOLOv11s(22)	0.33	8.00	13.48	26.75	97.82

Table 7. Comparison of effective receptive fields for YOLOv11s and DRR-YOLOv11s.

Table 7, the Max-Value of DRR-YOLOv11s improves significantly across all layers, with increases of 24.32% and 29.41% in the 10th and 16th layers, respectively. This demonstrates that DRR-YOLOv11s can effectively improve sensitivity to central features, and thus enhance the localization accuracy of small targets.

Further analysis of the ERF distribution reveals substantial optimization of multi-scale detection capability. For the layer dedicated to small target detection, namely layer 16, the areas of the high-contribution regions with thresholds of 20% and 30% increase by 0.25% and 0.45%, respectively, indicating an enhanced focus on core features. In the layer for medium target detection, layer 19, the area of the region with a 50% threshold grows by 1.74%, suggesting a balanced improvement between localized feature sensitivity and global perception. Notably, in the layer responsible for large target detection, layer 22, although the Max-Value remains stable, the area of the region with a 50% threshold increases by 3.44% while the area of the region with a 99% threshold slightly decreases. This demonstrates that DRR-YOLOv11s adapts to the detection requirements of large-scale targets by optimizing ERF distribution in deeper layers. Overall, these layer-specific improvements validate the effectiveness of DRR-YOLOv11s in enhancing multi-scale detection performance.

Figure 12 visualizes the changes in ERF distribution before and after model improvement. Subfigures (a)–(d) illustrate the effective receptive field distribution of the original YOLOv11s model at layers 10, 16, 19, and 22, respectively, while subfigures (e)–(h) depict the ERF characteristics of the improved DRR-YOLOv11s model at the corresponding layers. Comparative analysis indicates that the ERF ranges of the improved model expand to varying degrees across all layers.

This observation confirms that DRR-YOLOv11s effectively enhances the feature extraction capability for targets at different scales through a layer-specific ERF optimization strategy. Specifically, in the small target detection layer, localization accuracy is improved by strengthening the response to central features. In the medium target detection layer, feature representation is optimized by balancing local and global perception. For the large target detection layer, adaptation to large-scale target characteristics is achieved by adjusting the ERF

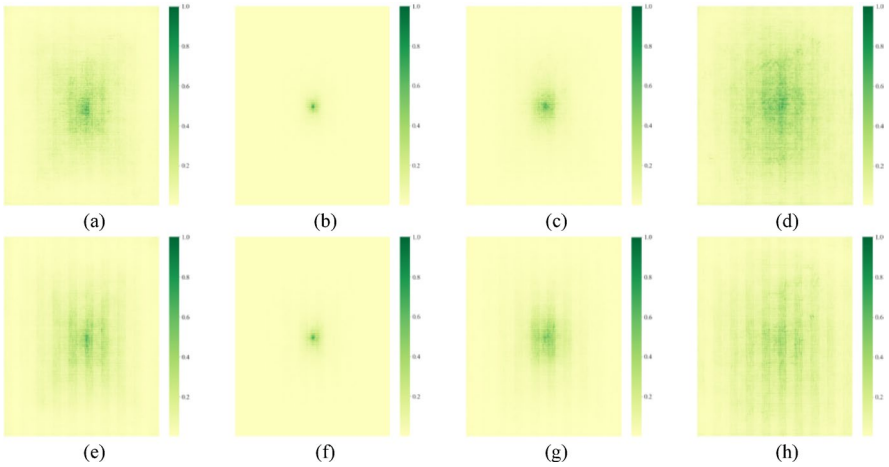


Fig. 12. Comparison of effective receptive fields of YOLOv11s and DRR-YOLOv11s.

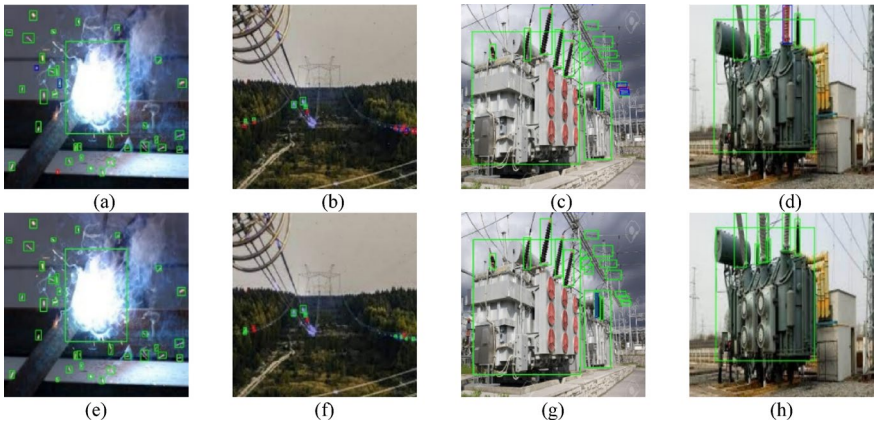


Fig. 13. Comparison of YOLOv11s and DRR-YOLOv11s detection results.

distribution range. This hierarchical ERF optimization strategy enables the improved model to handle multi-scale target detection tasks more efficiently and provides robust technical support for detection in complex scenes.

Target detection model evaluation and visualization

To more intuitively and quantitatively evaluate the correspondence between model predictions and actual labels, we visualize and analyze three types of detection results: False Positive (FP), False Negative (FN), and True Positive (TP). Figure 13 illustrates a comparison of detection results between the original YOLOv11s model and the improved DRR-YOLOv11s model.

Subfigures (a)–(d) present the detection results of the original YOLOv11s model, while subfigures (e)–(h) illustrate the results of the improved DRR-YOLOv11s model. In the visualization, green bounding boxes represent correct detections (TP), red bounding boxes indicate missed detections (FN), and blue bounding boxes correspond to false detections (FP).

By comparing subfigures (a) vs. (e) and (b) vs. (f), it is clearly evident that both models exhibit relatively good performance in detecting sparks, which are small targets. Notably, the DRR-YOLOv11s model shows a distinct advantage in identifying small spark targets. It can accurately detect instances that the original model tends to miss, thereby confirming the advantages of the improved model in detecting small targets. Further comparison of subfigures (c) vs. (g) and (d) vs. (h) shows that in the context of insulator small-target detection, the DRR-YOLOv11s model similarly attains superior performance compared to the original YOLOv11s model.

These visualization and comparative results indicate that DRR-YOLOv11s effectively enhances feature perception for small-scale targets through structural optimization. Consequently, the model significantly reduces the missed detection rate of tiny targets while maintaining high overall detection accuracy. Such improvements are particularly valuable for practical applications such as power equipment inspection, providing more reliable technical support for detecting minute defects in complex backgrounds.

Groups	RDSC	DDSConv	RVBs	Precision(%)	Recall(%)	mAP@0.5(%)	mAP@0.5:0.95(%)	Parameters(M)	GFLOPS(G)
G1				91.7	89.18	93.07	72.94	9.42	21.3
G2	√			92.42	90.07	94.28	74.00	9.38	20.9
G3		√		92.66	89.97	92.99	72.91	7.06	17.8
G4			√	92.10	89.36	92.70	73.09	9.57	24.3
G5	√	√		92.27	89.19	93.93	73.37	7.03	17.3
G6	√	√	√	92.50	89.42	93.46	73.45	5.88	15.2

Table 8. Comparison of ablation experiment results.

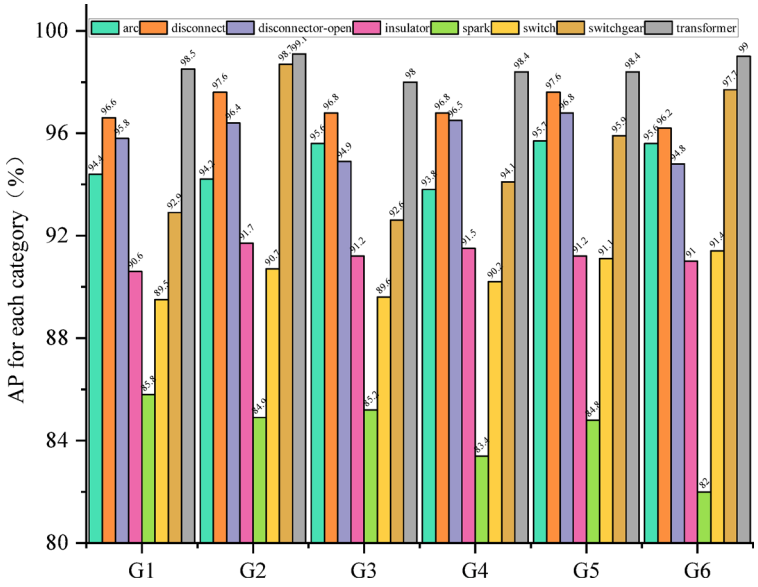


Fig. 14. Comparison of AP values for each category of targets in the ablation experiment.

Ablation experiment

To systematically assess the contributing factors of each module to the overall detection performance, this study designed six sets of elimination experiments. The experimental configurations are as follows: G1 represents the original YOLOv11s baseline model; G2 incorporates the RDSC module into the detection head; G3 replaces standard convolution with DDSConv fusion convolution; G4 substitutes the Bottleneck structure in C3K2 with the improved RepViTBlocks; G5 combines the G3 module with G2; and G6 integrates the G5 configuration with G4. The test results are presented in Table 8 and Fig. 14, where a "√" indicates that the corresponding enhancement strategy was applied.

As presented in Table 8, comparing G2 with G1 shows that incorporating the RDSC module alone lowers the parameter count by 0.04M and cuts computational complexity by 0.4G Simultaneously, the precision, recall, mAP@0.5, and mAP@0.5:0.95 increase by 0.72%, 0.89%, 1.21%, and 1.06%, respectively.

Comparing G3 with G1 demonstrates that incorporating DDSConv fusion convolutions reduces model parameters by 25.05% and computational complexity by 16.43%, while improving precision and recall by 0.96% and 0.79%, respectively. However, mAP@0.5 exhibits a slight decrease of 0.08%. Further analysis of G4 versus G1 shows that using the RepViTBlocks module alone increases model parameters by 0.15M and computational complexity by 3G, while precision and recall improve by 0.4% and 0.18%, respectively, and mAP@0.5 slightly decreases by 0.37%.

Subsequent experiments indicate that when RepViTBlocks is combined with other modules, it achieves performance improvements while reducing parameter count and computational complexity, demonstrating a clear synergistic optimization effect. Specifically, comparing G5 with G1, the combination of RDSC and DDSConv fusion convolution reduces parameters by 25.37% and computational complexity by 18.78%, while precision, recall, mAP@0.5, and mAP@0.5:0.95 improve by 0.57%, 0.01%, 0.86%, and 0.43%, respectively. In the case of G6 versus G1, the final improved DRR-YOLOv11s model achieves a 37.58% reduction in model parameters and a 28.64% reduction in computational complexity, with corresponding improvements in precision, recall, mAP@0.5, and mAP@0.5:0.95 of 0.8%, 0.24%, 0.39%, and 0.51%, respectively.

These results indicate that the proposed module combination can not only reduce model size and calculation costs but also improve overall detection performance, highlighting the effectiveness of the synergistic design strategy.

Figure 14 illustrates the detection performance across different target categories. The AP values of the arc, insulator, switch, switchgear, and transformer categories show marked improvement. Although slight decreases are observed in the AP values of the disconnector, disconnector_open, and spark categories, the overall detection performance remains satisfactory, particularly given the substantial reductions in model parameters and computational complexity.

These results demonstrate that DRR-YOLOv11s achieves consistent improvements in detection performance while significantly reducing model complexity through a multi-module collaborative optimization strategy. The proposed hierarchical optimization design introduces a novel method for developing efficient target detection models, making it suitable for real-time applications under computational constraints.

Comparison experiment on fault detection of power equipment

Under a unified experimental setup, this study systematically compares the detection performance of multiple mainstream target detection models using the same dataset. Table 9 presents the quantitative evaluation results for each model in the task of power equipment fault detection. The experimental results indicate that the proposed DRR-YOLOv11s model exhibits clear advantages in this task. Specifically, the model achieves an accuracy of 92.50%, a recall of 89.42%, an mAP@0.5 of 93.46%, and an mAP@0.5:0.95 of 73.45%. It operates at a speed of 160.22 frames per second, features 5.88 million parameters, and possesses a computational complexity of 15.2 GFLOPs, all of which place it at a leading performance level.

Compared with the YOLOv5s model, the proposed DRR-YOLOv11s model achieves a 35.46% reduction in parameters and a 36.13% reduction in computational cost, while improving Precision, mAP@0.5, and mAP@0.5:0.95 by 3.54%, 0.97%, and 3.31%, respectively. Although the detection speed slightly decreases by 9.94 FPS and Recall drops by 0.38%, the overall performance remains superior. Compared with YOLOv8s, the proposed model reduces parameters and computational cost by 47.17% and 46.67%, respectively. The detection speed slightly decreases by 1.71 FPS, while Precision and mAP@0.5:0.95 increase by 0.23% and 1.35%, and Recall and mAP@0.5 slightly decrease by 0.78% and 0.39%, maintaining an overall performance advantage.

Compared with YOLOv9s, the parameters and computation are reduced by 17.30% and 43.07%, respectively, and the detection speed is significantly improved by 64.63 FPS. Moreover, Precision, Recall, mAP@0.5, and mAP@0.5:0.95 all exhibit improvements. Compared with YOLOv10s, the parameters and computational cost are reduced by 26.87% and 37.96%, respectively, and the detection speed increases by 24.58 FPS. Both Precision and Recall show slight improvements, while mAP@0.5 and mAP@0.5:0.95 decrease marginally by 0.05% and 0.4%, respectively. Compared with YOLOv12s, the parameters and computational cost are reduced by 35.24% and 21.24%, respectively. The proposed model outperforms.

YOLOv12s across all metrics, achieving gains of 58.36 FPS, 1.31% Precision, 1.8% Recall, 1.21% mAP@0.5, and 1.99% mAP@0.5:0.95. Compared with the baseline YOLOv11s model, the parameters and computational cost are reduced by 37.58% and 28.64%, respectively. The detection speed slightly increases by 0.37 FPS, while Precision, Recall, mAP@0.5, and mAP@0.5:0.95 improve by 0.8%, 0.24%, 0.39%, and 0.51%, respectively.

In summary, the proposed DRR-YOLOv11s algorithm achieves a substantial reduction in model size and computational complexity while maintaining or slightly enhancing detection speed, precision, recall, and mean average precision. Compared with other models, DRR-YOLOv11s demonstrates superior balance between efficiency and accuracy, making it highly suitable for resource-constrained deployment environments with strong engineering applicability and practical value.

Comparison experiment of personal protective equipment detection

To evaluate the effectiveness of different lightweight strategies for power equipment fault detection, we conducted comparative experiments under the same input resolution (640 × 640) and consistent training settings, as shown in Table 10. The compared methods include lightweight backbone replacement (MobileNetv3/EfficientNetv2/ShuffleNetv2) and our proposed collaborative structural lightweight design (DDSConv + RDSC + RVBs).

Experimental results show that simply replacing the backbone with lightweight architectures can significantly increase inference speed or reduce computational cost, but it often leads to degraded accuracy. For instance, YOLOv11s-EfficientNetv2 and YOLO11s-ShuffleNetv2 achieve inference speeds of 221.58 FPS and 266.34 FPS, respectively; however, compared with the YOLOv11s baseline, their Precision/Recall/mAP@0.5 drop by 5.85%/7.24%/6.89% (EfficientNetv2) and 0.48%/2.56%/2.81% (ShuffleNetv2). This indicates that, in power-equipment fault detection scenarios characterized by small targets, weak textures, and heavy background interference, overly compressed backbones may provide insufficient feature representation, resulting in more

Detection Model	Precision(%)	Recall(%)	mAP@0.5(%)	mAP@0.5:0.95 (%)	FPS	Params (M)	GFLOPs(G)
YOLOv5s	88.96	89.80	92.49	70.14	170.14	9.11	23.8
YOLOv8s	92.27	90.20	93.85	72.10	161.91	11.13	28.5
YOLOv9s	91.58	89.08	93.33	73.50	95.57	7.17	26.7
YOLOv10s	92.44	88.78	93.51	73.85	135.62	8.04	24.5
YOLOv12s	91.19	87.62	92.19	71.46	101.84	9.08	19.3
YOLOv11s	91.70	89.18	93.07	72.94	159.83	9.42	21.3
DRR-YOLOv11s	92.50	89.42	93.46	73.45	160.22	5.88	15.2

Table 9. Comparison experimental results of multiple mainstream target detection models in power equipment fault detection task.

Detection Model	Precision(%)	Recall(%)	mAP@0.5(%)	mAP@0.5:0.95(%)	FPS	Params(M)	GFLOPs(G)
YOLOv11s-MobileNetv3	90.98	86.34	91.51	67.50	157.09	4.57	7.6
YOLOv11s-EfficientNetv2	85.85	81.94	86.18	62.10	221.58	8.56	8.7
YOLO11s-ShuffleNetv2	91.22	86.62	90.26	68.29	266.34	5.33	10.5
YOLOv11s	91.70	89.18	93.07	72.94	159.83	9.42	21.3
DRR-YOLOv11s	92.50	89.42	93.46	73.45	160.22	5.88	15.2

Table 10. Comparative Results of Common Lightweight Methods for Power Equipment Fault Detection.

	Average Precision AP (%)								
Detection model	Gloves	Helmet	Person	Shoe	Vest	Precision(%)	Recall(%)	mAP@0.5(%)	mAP@0.5:0.95(%)
YOLOv5s	91.7	96.3	95.4	90.4	97.9	92.47	90.55	94.33	77.23
YOLOv8s	92.3	96.2	95.5	91	98.1	93.85	89.98	94.62	78.30
YOLOv9s	92.8	95.8	95.5	90.9	98	93.16	90.17	94.58	78.30
YOLOv10s	91.9	95.9	95.7	91.1	97.6	93.59	89.46	94.46	78.59
YOLOv12s	92	96	95.8	90.4	98	93.17	90.06	94.44	77.28
YOLOv11s	91.7	95.5	95.9	91.5	98.2	94.15	89.91	94.55	78.15
DRR-YOLOv11s	91.7	95.9	95.6	90.8	98.1	93.32	90.32	94.52	77.72

Table 11. Comparative experimental results of multiple mainstream target detection models in the PPE detection task.

missed detections and unstable localization. Meanwhile, YOLOv11s-MobileNetv3 has the lowest parameter count and computation (4.57 M / 7.6 GFLOPs), but its overall performance still lags behind the baseline, with Precision/Recall/mAP@0.5 decreasing by 0.72%/2.84%/1.56%, respectively.

In contrast, DRR-YOLOv11s achieves the best accuracy–efficiency trade-off while maintaining real-time performance. Compared with the baseline YOLOv11s, DRR-YOLOv11s reduces the parameter count by 37.58% and the computational cost by 28.64%, while improving Precision, Recall, and mAP@0.5 by 0.80%, 0.24%, and 0.39%, respectively. This indicates that our lightweight design is not a single-point compression achieved by "backbone replacement"; instead, it is a coordinated strategy that combines system-level lightweighting in the backbone/neck, small-object compensation in the detection head, and deployment acceleration via structural re-parameterization, thereby reducing computation while preserving—and even enhancing—the discriminative capability for small fault targets.

Comparison experiment of personal protective equipment detection

The improved DRR-YOLOv11s model presented in this study demonstrates robust detection properties on the Personal Protective Equipment (PPE) dataset, which contains images of safety equipment worn by workers. In Table 11, the model demonstrates balanced properties in different categories of protective equipment, including gloves, helmets, people, shoes, and vests. Although its detection performance on the PPE dataset is slightly lower than that on the Electrical Equipment Failure (EEF) dataset, the model still attains results comparable or superior to those of other benchmark models in several performance metrics. These findings indicate that the proposed optimization strategy enhances detection capability in specialized domains while maintaining the model's generalization ability, making it highly suitable for multi-scenario industrial safety monitoring applications.

Demonstration of DRR-YOLOv11s detection results

To validate the performance advantages of the DRR-YOLOv11s model, comparative experiments were carried out on both the Electrical Equipment Failure (EEF) dataset and the Personal Protective Equipment (PPE) dataset. The prediction results under the same experimental configuration are visually presented in Fig. 15. The results indicate that the proposed DRR-YOLOv11s model achieves outstanding performance in both industrial inspection tasks, with detection accuracy and reliability comparable to those of professional manual inspections. These findings demonstrate that DRR-YOLOv11s effectively achieves a synergistic optimization of detection precision and computational efficiency through multi-scale feature fusion and a lightweight network design, providing an efficient and reliable solution for industrial intelligent monitoring applications.

Discussion

The improved performance of DRR-YOLOv11s can be attributed to a collaborative, multi-module lightweight design rather than a simple stacking of existing techniques. The ablation study (Table 8) suggests that each module contributes differently and that their combination yields complementary effects. Specifically, introducing RDSC into the decoupled detection head improves precision and recall and increases mAP, indicating enhanced localization and classification capability at the head stage (G2 vs. G1). In contrast, replacing standard convolutions with DDSConv effectively reduces parameters and FLOPs while keeping precision and recall competitive

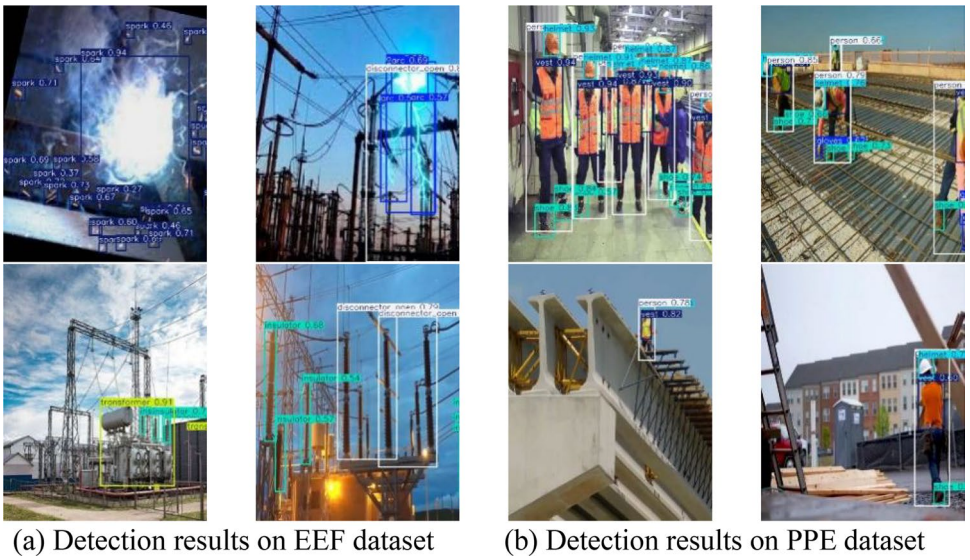


Fig. 15. Test results on different datasets.The images are derived from publicly available datasets on the Roboflow platform.

(G3 vs. G1), suggesting that the proposed lightweight feature extraction can reduce redundancy with limited accuracy loss. Although RVBs alone do not reduce computation in isolation (G4 vs. G1), they strengthen feature interaction and become more effective when combined with DDSConv and RDSC, resulting in a favorable overall trade-off in G6 (92.50% precision, 89.42% recall, 93.46% mAP@0.5, 73.45% mAP@0.5:0.95, with only 5.88 M parameters and 15.2 GFLOPs).

Visualization-based interpretation

To better understand why DRR-YOLOv11s improves detection, we provide qualitative and quantitative visual analyses. First, the effective receptive field (ERF) comparison (Fig. 12) shows that the improved model expands ERF across layers and adapts the ERF distribution for different detection scales, strengthening central responses for small objects, balancing local/global perception for medium objects, and adjusting the ERF range for large objects. Second, the FP/FN/TP visualization (Fig. 13) suggests that DRR-YOLOv11s reduces missed detections on tiny targets compared with the baseline, indicating that head-level enhancement helps compensate for the representational limitations introduced by lightweight backbone operations.

Speed–accuracy and small–large object trade-offs

Industrial monitoring in hydropower plants requires both real-time inference and reliable recognition of small fault patterns. DRR-YOLOv11s achieves a relatively balanced trade-off by combining front-end cost reduction (DDSConv+RVBs) with head-level compensation (RDSC). Overall, it reduces parameters by 37.58% and computation by 28.64% while reaching 160.22 FPS and maintaining competitive detection accuracy on EEF. The results indicate improved performance on small objects, while ERF analysis supports stable performance across multi-scale targets.

Failure cases, bias, and limitations

Despite these results, several limitations remain. Both datasets may exhibit class imbalance and long-tail categories, which may bias training toward dominant classes. Small or visually ambiguous objects are sensitive to annotation quality, and incomplete labels may be counted as false positives. Dataset characteristics such as correlated samples may affect evaluation difficulty and generalization. The cross-dataset evaluation retrains the model under consistent settings, which helps assess stability but does not fully represent zero-shot transfer capability.

Conclusion

To meet the stringent requirements of real-time intelligent monitoring in hydropower plant O&M scenarios, this paper proposes a multi-dimensional lightweight detection algorithm, DRR-YOLOv11s, based on YOLOv11s. The proposed method integrates three complementary modules: DDSConv for low-cost yet expressive feature extraction, RDSC as a detection-head enhancement module to improve sensitivity to small objects, and RVBs featuring efficient feature interaction and deployment-friendly design. Ablation studies suggest that these modules provide complementary benefits in improving detection accuracy while reducing computational overhead, and their combined use achieves a favorable overall performance.

On the EEF dataset, DRR-YOLOv11s achieves 92.50% precision, 89.42% recall, 93.46% mAP@0.5, and 73.45% mAP@0.5:0.95, while reducing model complexity to 5.88M parameters and 15.2 GFLOPs and reaching an inference speed of 160.22 FPS, indicating its potential feasibility for edge deployment. Cross-dataset validation

on the PPE dataset further suggests that this lightweight design remains stable and effective across different target categories and imaging conditions. Overall, DRR-YOLOv11s provides a promising solution that balances accuracy and efficiency for equipment-fault detection and worker PPE recognition in complex hydropower scenarios, offering a practical accuracy–computation trade-off for industrial safety monitoring.

Future work will expand multi-site, multi-season real-world data and improve annotation quality, introduce stronger long-tail training strategies (e.g., re-weighting, re-sampling, and class-aware augmentation) to enhance rare-fault recognition, and explore deployment-oriented optimizations (e.g., quantization, pruning, and energy-aware dynamic inference) to further improve robustness and efficiency on edge devices.

Data availability

The two publicly available datasets used in this study, Electrical Equipment Failure (EEF) and Personal Protective Equipment (PPE), are both provided by the Roboflow platform. They can be accessed at https://universe.roboflow.com/ppe-detection-7kco8/station-raaga-ogkgf/dataset/2 and https://universe.roboflow.com/liu-hxoap/yui-2s6gc-180su/dataset/2, respectively. Given the relatively limited sample size of the EEF dataset, image augmentation techniques built into the Roboflow platform were applied to effectively expand its training set. The augmented EEF dataset, together with the original (non-augmented) PPE dataset, has been publicly shared and is available at https://github.com/zongbaoshu/Electrical-Equipment-Failure-Dataset-and-Personal-Protective-Equipment-Dataset. The codes used in this study are available from the corresponding author upon reasonable request.

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Author contributions

P.Y. and B.Z. designed the research framework and proposed the DRR-YOLOv11s model. X.G. and X.X. were responsible for model implementation and conducting the experiments. S.C. and Y.L. were in charge of data collection and preprocessing, including the construction of the Electrical Equipment Failure and Personal Protective Equipment datasets. P.Y. and X.G. performed performance analysis and ablation studies. B.Z. wrote the main manuscript text, while S.C. and Y.L. prepared Figs. 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14 and 15 and Tables 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 and 11. All authors contributed to the manuscript revision and reviewed and approved the final version for submission.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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